

A Universal Finite-Response Law for Energy Storage and Release in Driven Physical Systems

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A wide range of physical systems exhibit delayed response, hysteresis, and non-instantaneous energy release under external forcing. These phenomena appear across condensed matter systems, superfluids, plasmas, and strongly driven collapse processes such as sonoluminescence. Despite their ubiquity, these effects are typically treated as system-specific rather than manifestations of a common underlying principle.

In this work, we propose a universal finite-response law governing the evolution of an internal state variable that mediates energy storage and release. We show that energy release is controlled not by instantaneous forcing or geometric extrema, but by the dynamics of this delayed internal state. This framework naturally explains lag, hysteresis, and phase shifts observed across diverse systems.

A key prediction of the model is a transition in energy-release timing as the forcing timescale approaches the intrinsic response timescale. We propose experimental tests based on controlled forcing sweeps that can distinguish finite-response dynamics from instantaneous-response models. The framework provides a unifying description of non-equilibrium response and suggests a deeper medium-based interpretation of physical dynamics.

I. INTRODUCTION

Many physical systems exhibit behaviour that cannot be fully explained by instantaneous-response models. Under external forcing, systems often display lag, hysteresis, delayed emission, and non-equilibrium dynamics. These features are well documented in condensed matter systems, viscoelastic media, superfluids, plasmas, and driven collapse phenomena such as sonoluminescence.

Despite their ubiquity, these effects are typically treated as system-specific. There is currently no unified framework that identifies a common dynamical structure governing energy storage and release across these domains.

In this work, we propose that these phenomena are manifestations of a universal finite-response law. We argue that energy release is governed by the dynamics of an internal state variable with a finite response timescale, rather than by instantaneous forcing or geometric extrema.

II. FINITE-RESPONSE LAW

We introduce an internal state variable $Q(t)$ that evolves according to

$$\tau_Q \frac{dQ}{dt} = Q_{\text{eq}}(F) - Q, \quad (1)$$

where $F(t)$ represents the external forcing, $Q_{\text{eq}}(F)$ is the equilibrium state associated with that forcing, and τ_Q is the intrinsic response timescale.

Energy release is assumed to occur when the system exceeds a threshold:

$$L_{\text{rel}}(t) \propto \sigma(Q - Q_{\text{crit}}) \cdot |\dot{Q}|, \quad (2)$$

where σ is a smooth activation function.

This formulation implies that observable release is governed by the internal dynamics of Q , not directly by F .

III. CROSS-DISCIPLINARY EVIDENCE

A. Condensed Matter Systems

Hysteresis in magnetic and ferroelectric systems demonstrates that internal state variables lag applied fields. Similarly, viscoelastic materials exhibit delayed stress-strain response, indicating finite response times.

B. Superfluids

In superfluid helium, heat propagates as second sound, demonstrating that energy transport occurs via wave-like dynamics with finite propagation speed. This behaviour reflects delayed internal response and multi-component coupling.

C. Plasma and Collapse Systems

In plasma pinches and collapse phenomena, energy release often occurs after peak compression, indicating that emission is governed by internal redistribution processes rather than geometric extrema alone.

D. Sonoluminescence

Driven bubble collapse provides a particularly clean example. Numerical and experimental studies indicate that emission timing is not strictly locked to minimum radius. Instead, it reflects internal state evolution, including delayed coherence or energy redistribution processes.

IV. KEY PREDICTION: TIMING TRANSITION

A central prediction of the finite-response framework is that energy release timing depends on the ratio of forcing and response timescales.

Define

$$\Delta t_{\text{rel}} = t_{\text{rel,max}} - t_{F,\text{max}}. \quad (3)$$

Then:

- For $\tau_Q \ll \tau_{\text{force}}$: $\Delta t_{\text{rel}} \approx 0$
- For $\tau_Q \sim \tau_{\text{force}}$: $\Delta t_{\text{rel}} > 0$
- For $\tau_Q \gg \tau_{\text{force}}$: release follows internal state, not forcing

This predicts a transition from geometry-locked to response-locked behaviour.

V. PROPOSED EXPERIMENTAL TEST

We propose a controlled forcing experiment:

- Apply periodic forcing with variable frequency
- Measure forcing peak time t_F

- Measure energy release peak t_{rel}

The key observable is

$$\Delta t_{\text{rel}} = f\left(\frac{\tau_Q}{\tau_{\text{force}}}\right). \quad (4)$$

A non-zero, frequency-dependent lag would provide direct evidence of finite-response dynamics.

A. Figure Placeholder

VI. RELATION TO THERMODYNAMICS

In this framework, temperature is interpreted as a derived quantity associated with redistribution of stored internal energy during response. Irreversibility arises naturally from finite response times and delayed equilibration.

VII. INTERPRETATION

One possible interpretation is that this universal response behaviour reflects an underlying medium with finite coherence and response timescales. In this view, diverse physical systems probe different regimes of the same underlying response dynamics.

VIII. CONCLUSION

We have proposed a universal finite-response law governing energy storage and release in driven systems. The framework unifies lag, hysteresis, and delayed emission across multiple domains and predicts a measurable transition in release timing.

Future experimental tests can determine whether this response law represents a fundamental principle of physical dynamics.

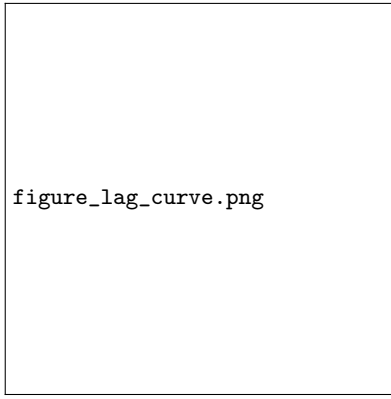


FIG. 1. Predicted lag between forcing and release as a function of timescale ratio.